SCIENCE
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TECHNOLOGY

Near Field Electromagnetic Interference Coupling-Mechanisms of Multilayer Ceramic Capacitors

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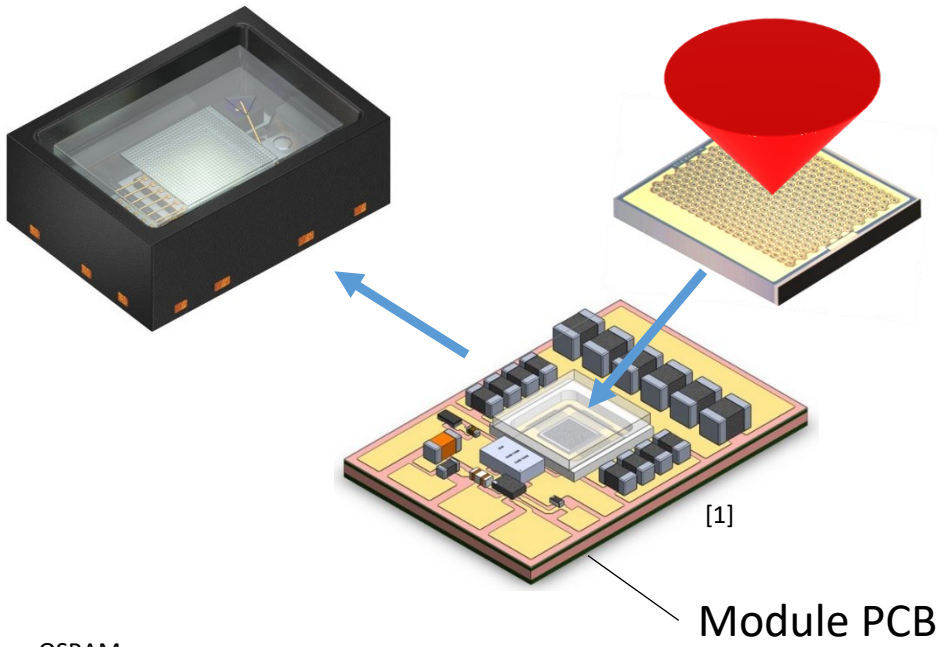
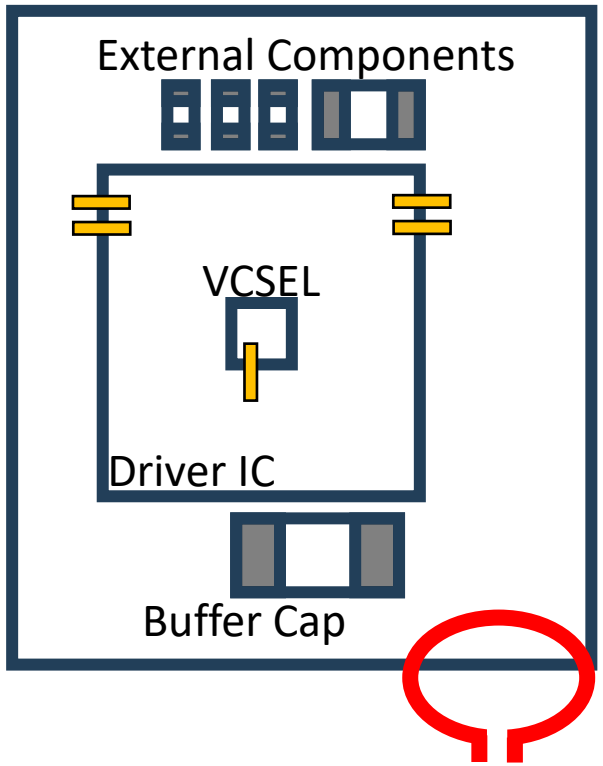
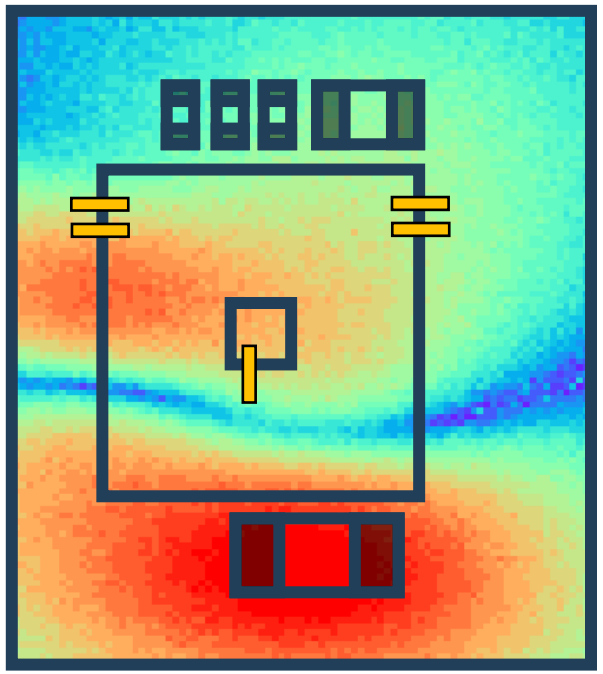
Application, Objective

- **V**ertical **C**avity **S**urface **E**mitting **L**asers are used for all kind of structured light applications, e.g. LIDAR, ToF, etc.
- They need high, fast pulsing currents conducted on bond wires for operation (rise time: hundreds of pico sec.)
- Buffer capacitors used for spike current demand







[1] Vixar, ams OSRAM

Outline

- Introduction
  - TEM cell and IC stripline method
- DUT & Measurement Setup
- Results
  - Coupling coefficients
  - Equivalent extracted dipoles
  - Field plots standing wave
- Conclusion & Outlook

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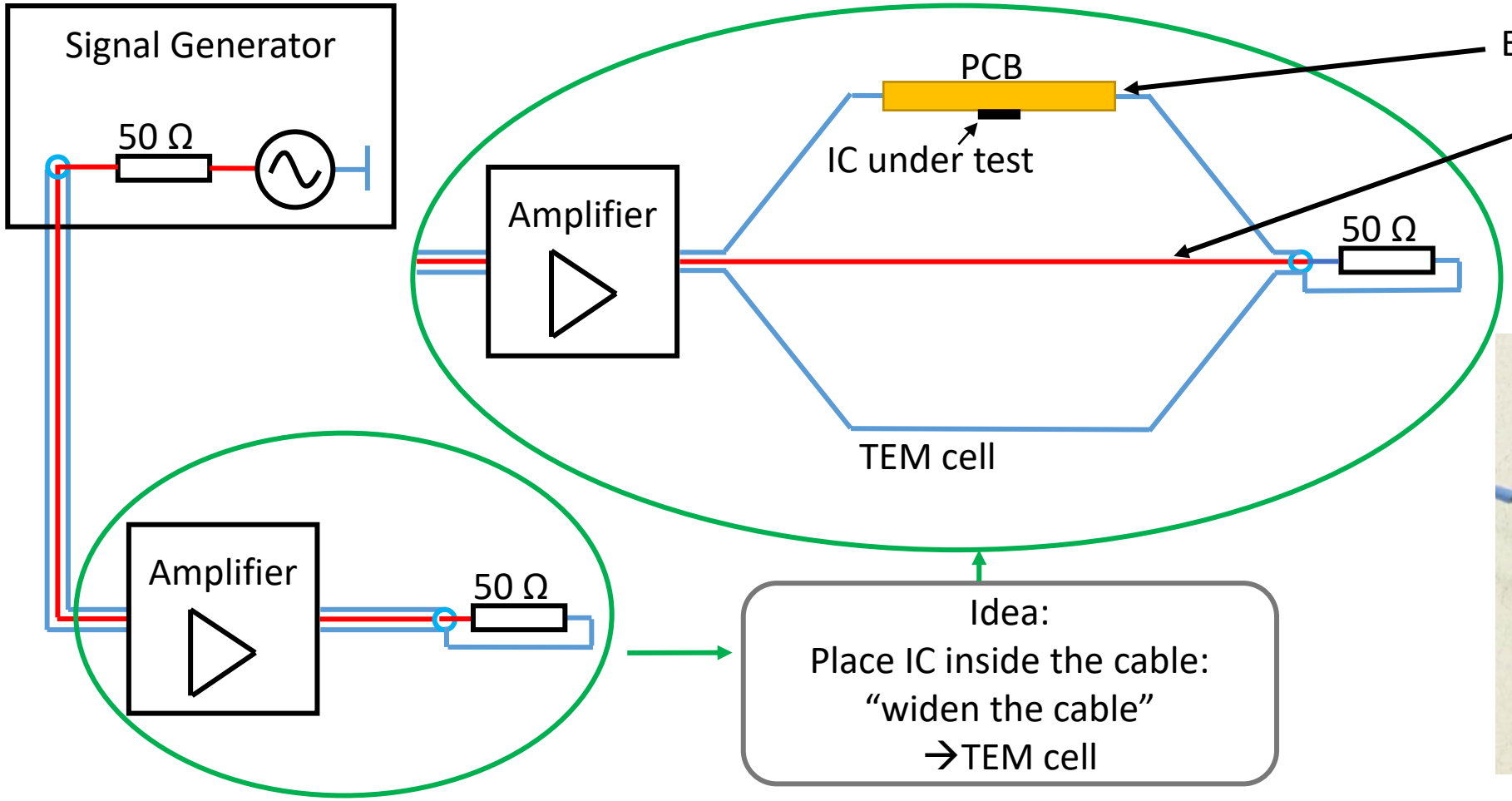
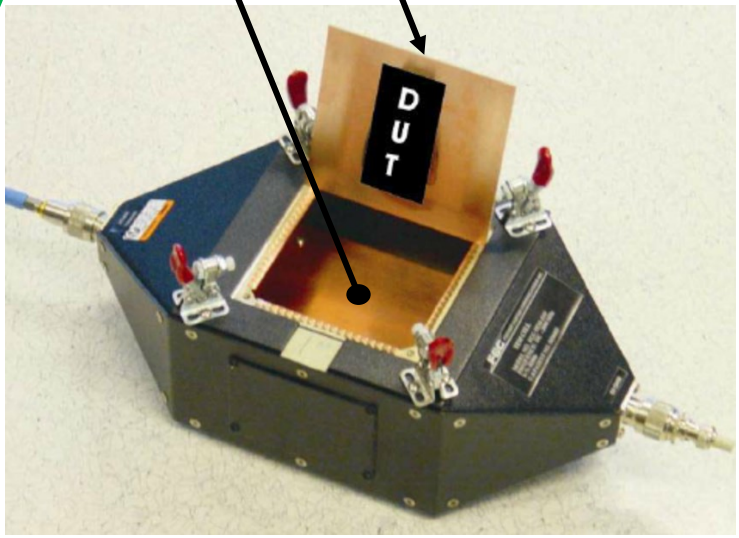
TEM cell method (IEC 61967 & IEC 62132)





Idea:
Place IC inside the cable:
“widen the cable”
→TEM cell

[2]

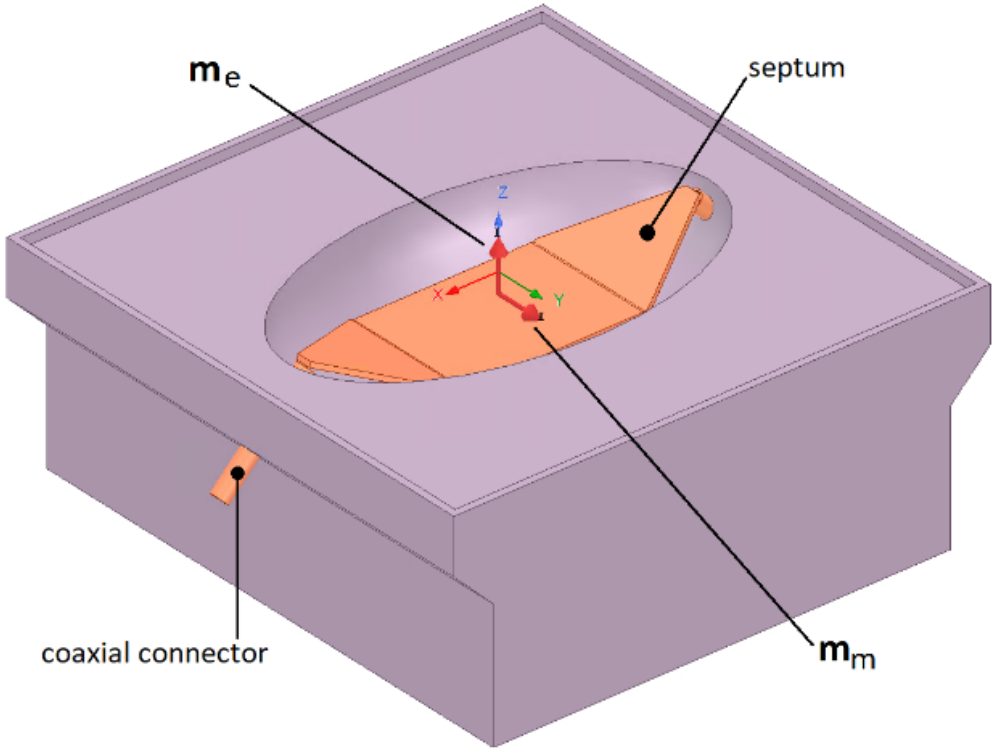
[2] T. Ostermann and B. Deutschmann. "TEM-cell and surface scan to identify the electromagnetic emission of integrated circuits." Proceedings of the 13th ACM Great Lakes symposium on VLSI. 2003.

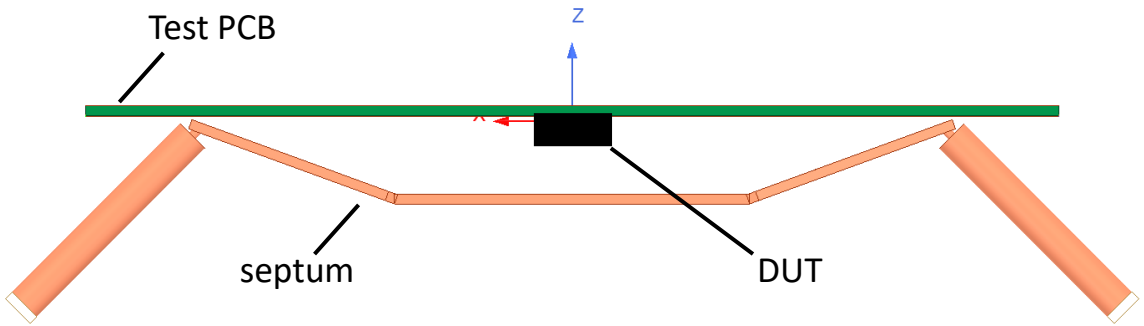
IC stripline method

- Smaller DUT-septum distance → higher field strengths, up to 10 kV/m for immunity tests
- Sources that can excite a propagating mode inside a TEM waveguide:

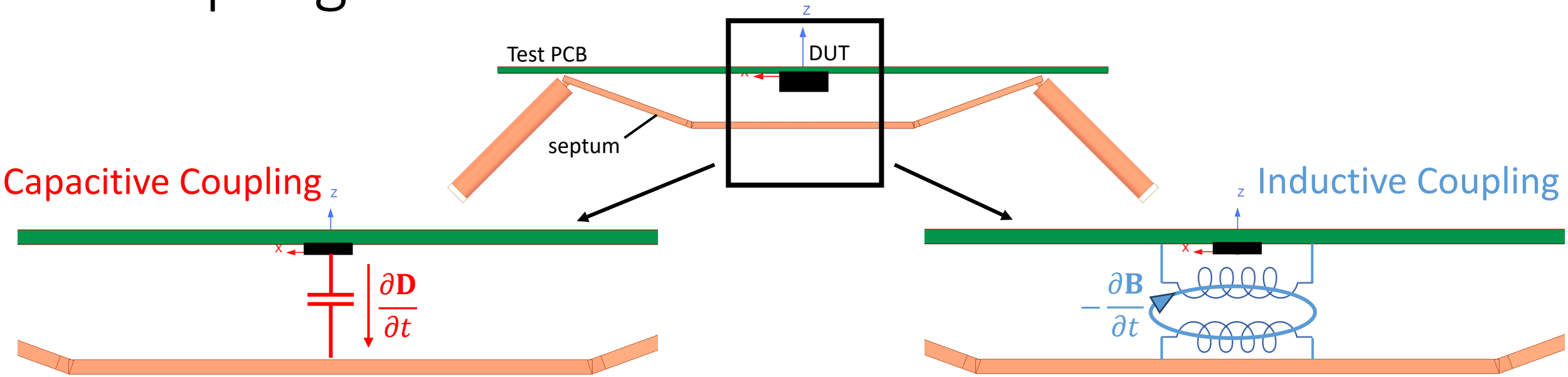
  Electric Hertzian dipole moment $\mathbf{m_e}$ in z-direction

  Magnetic Hertzian dipole moment $\mathbf{m_m}$ in y-direction
- Sources in other directions will not excite a propagating wave

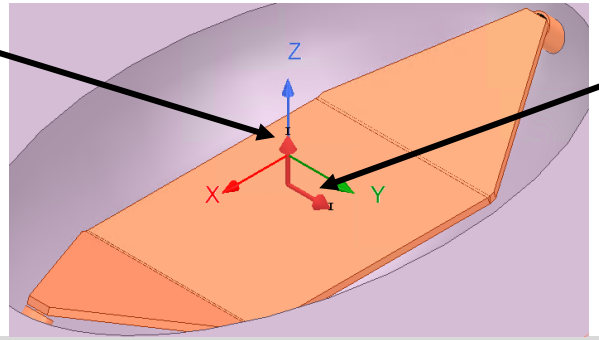




Coupling mechanisms



Equivalent electric dipole moment in z direction



Equivalent magnetic dipole moment in y direction

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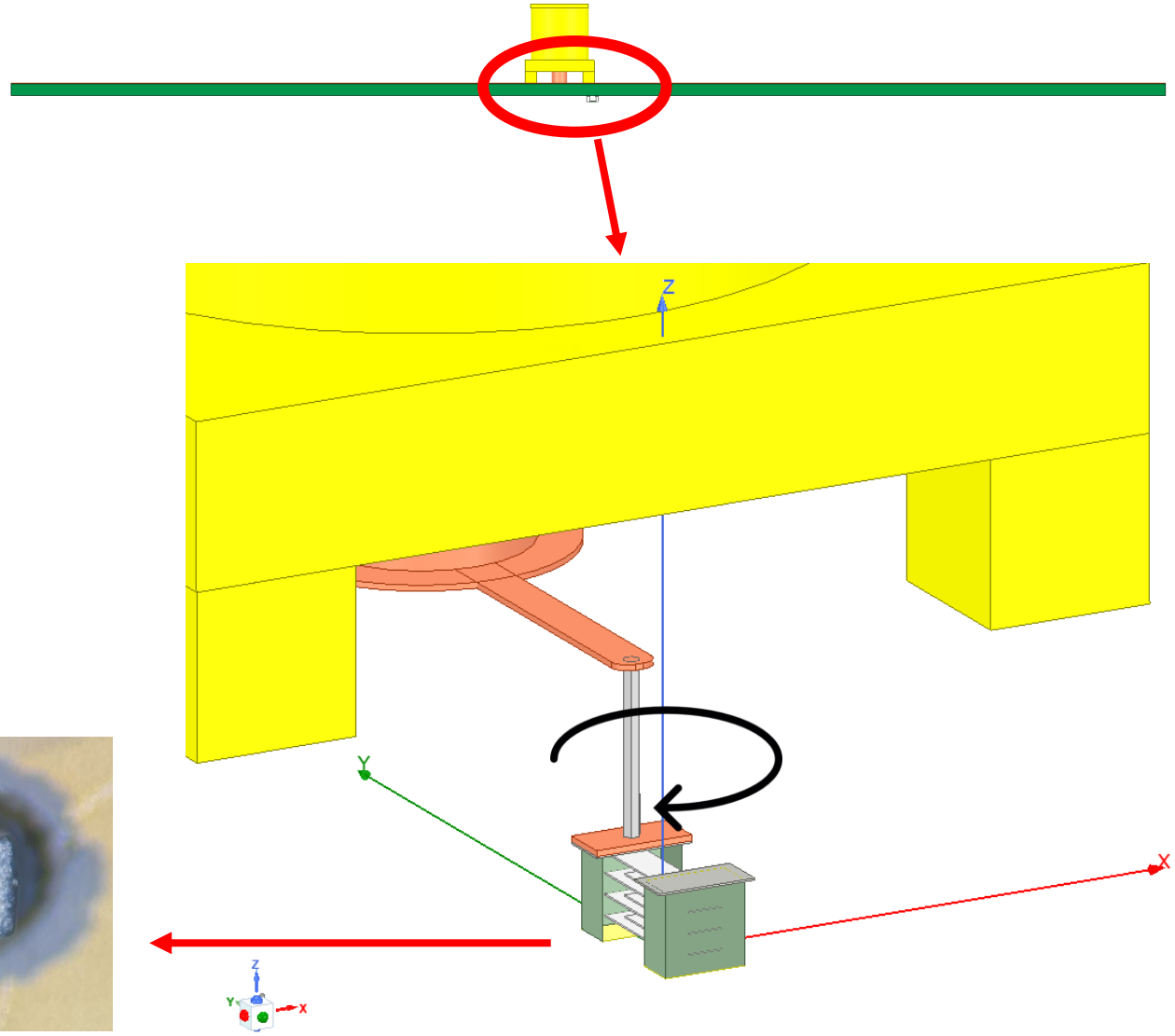
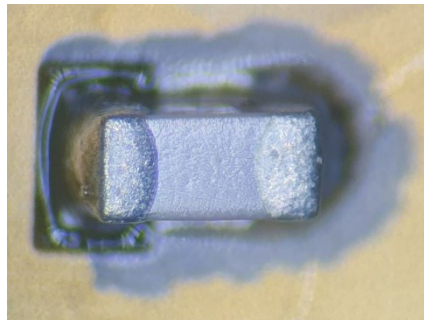
MLCC on Test PCB

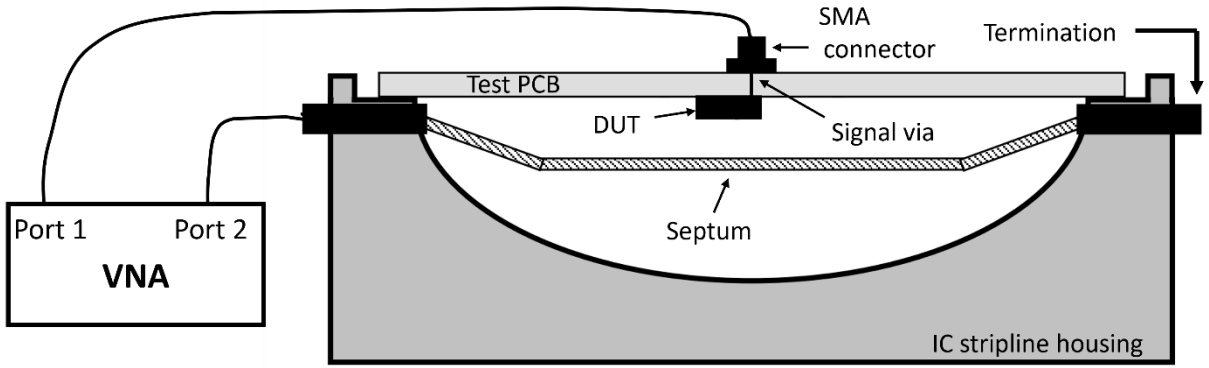


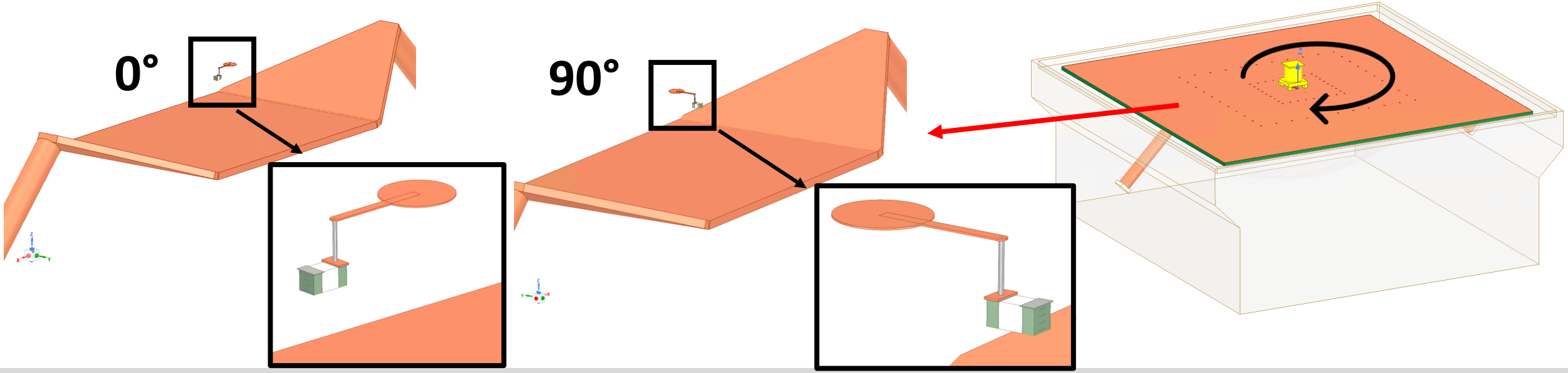
- 0402 (imperial) 9 pF MLCC capacitor
- Soldered onto standard conform 100 x 100 mm test PCB from a signal pad to GND plane
- EM FEM model with exact internal geometry was modelled
- Soldered onto PCB 2x → rotated around z-axis



Test Setup

- Test PCB inserted into IC stripline
- Vector network analyzer used to extract transmission coefficient (Scattering parameter S21)
- Measured at 0 and 90° PCB rotation angle





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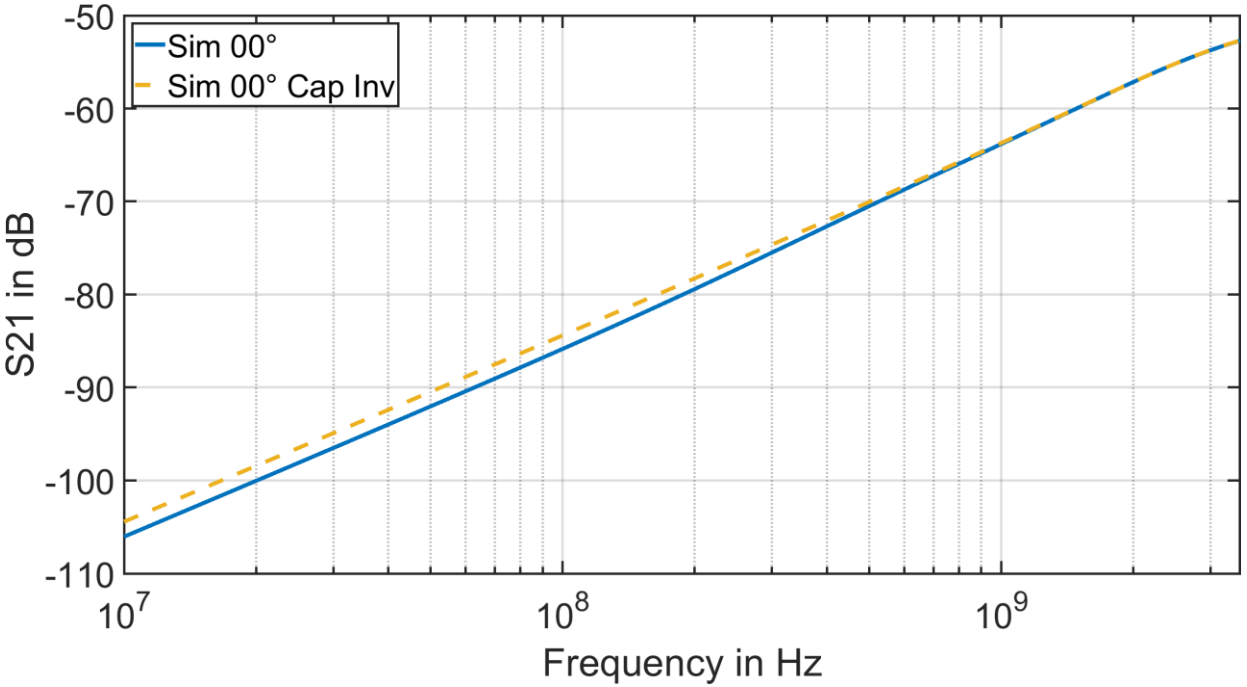
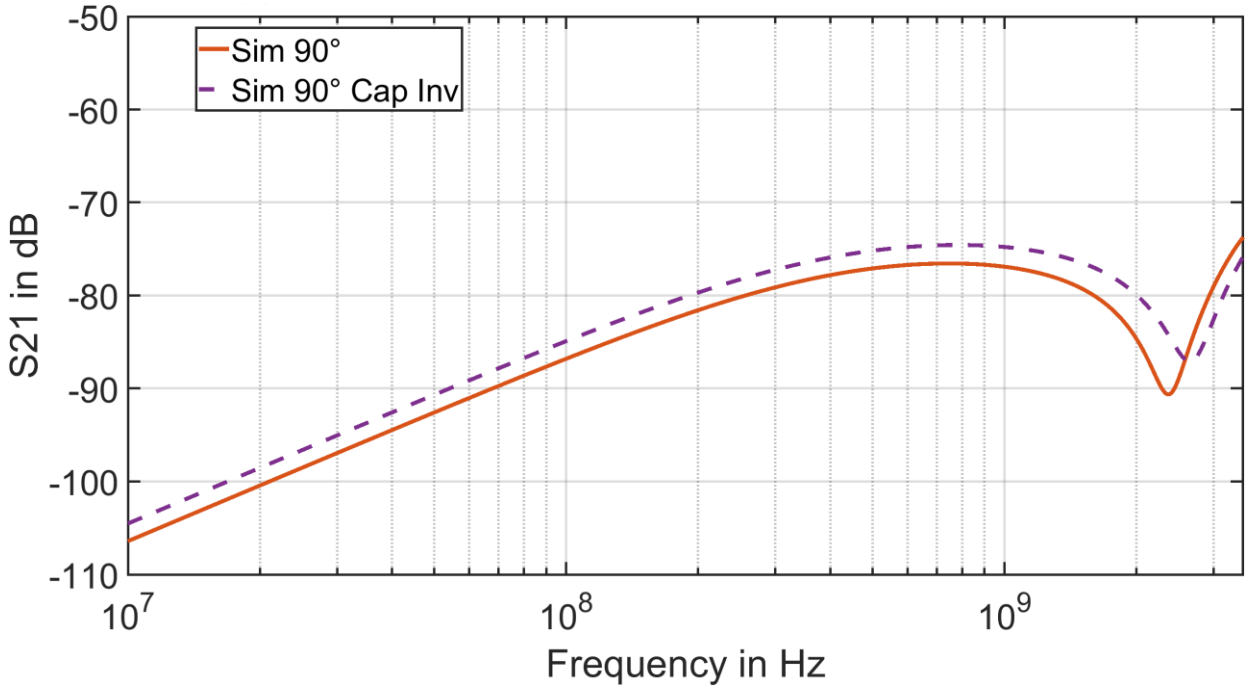
Coupling factors S21



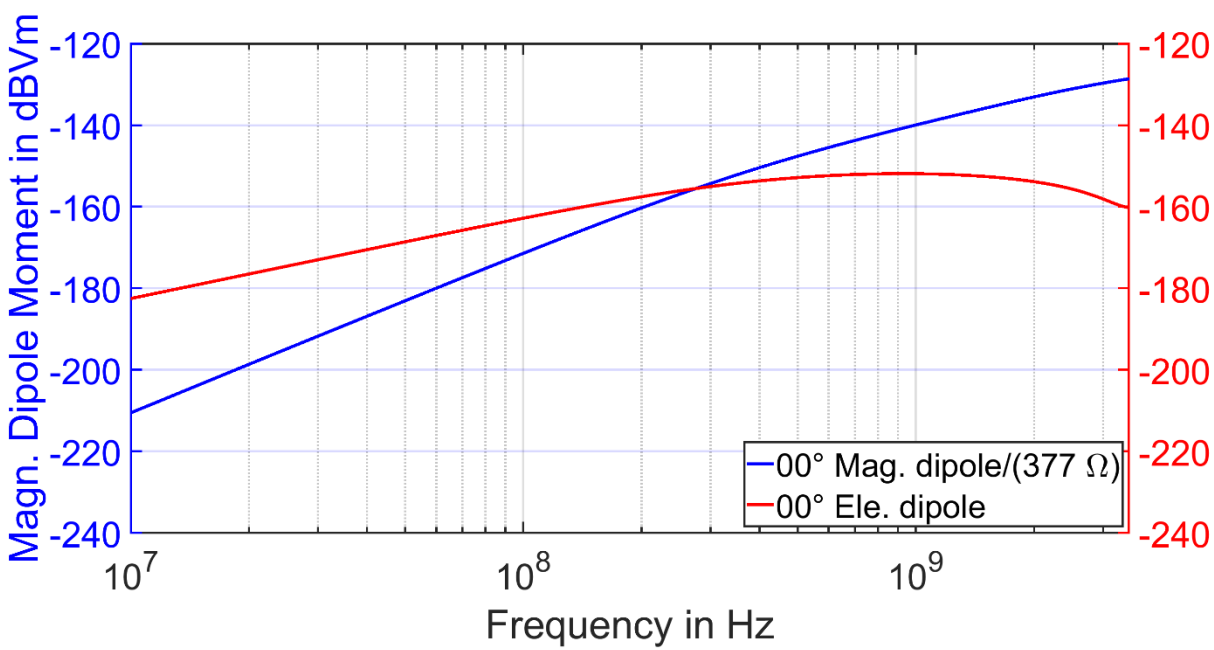


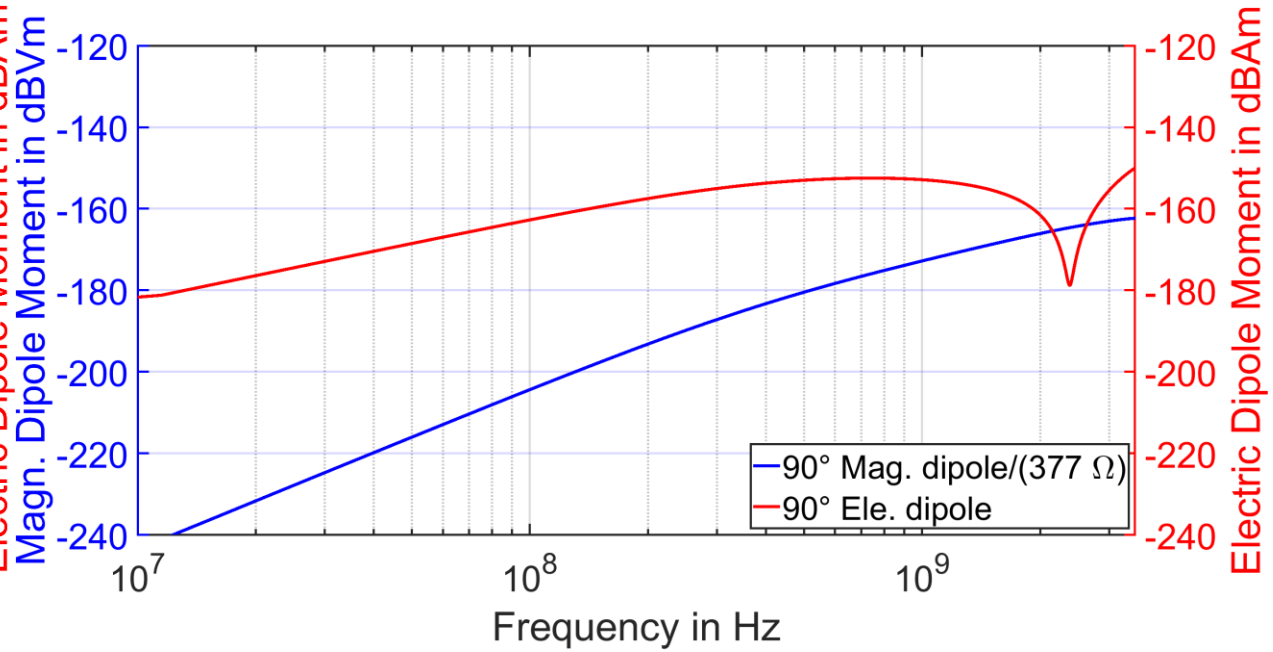
Inverted

Not Inverted

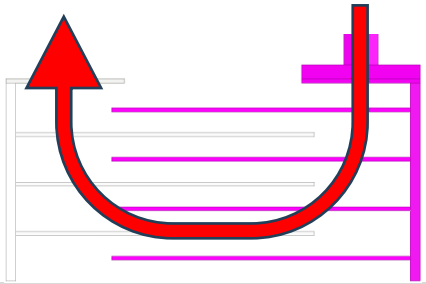
- Shielding Effect of GND fingers
- Why not in 0° case higher frequency?
- Local minimum in 90° case

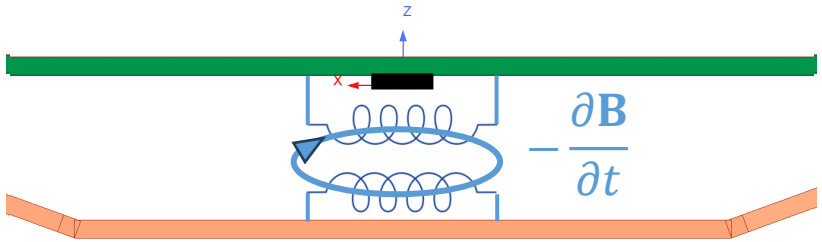
Equivalent Hertzian Dipole Moments



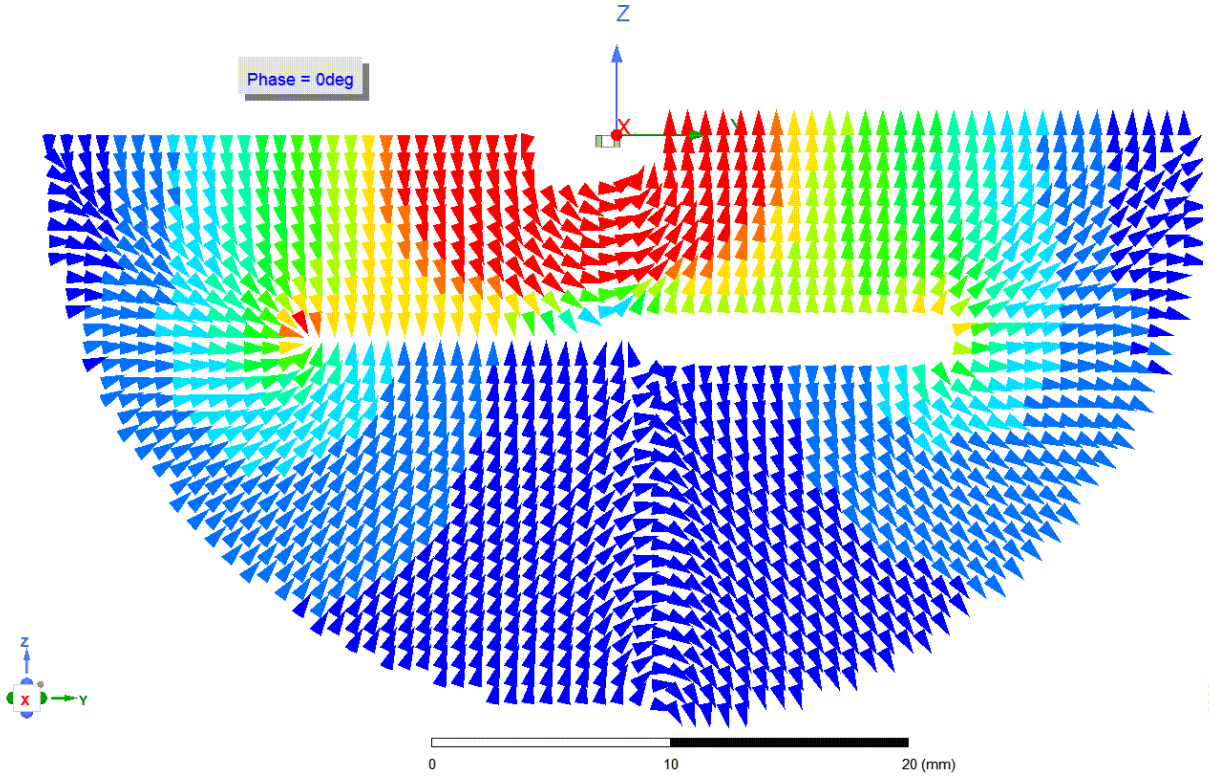


$\frac{\partial \mathbf{D}}{\partial t} \rightarrow j\omega\varepsilon\mathbf{E}$

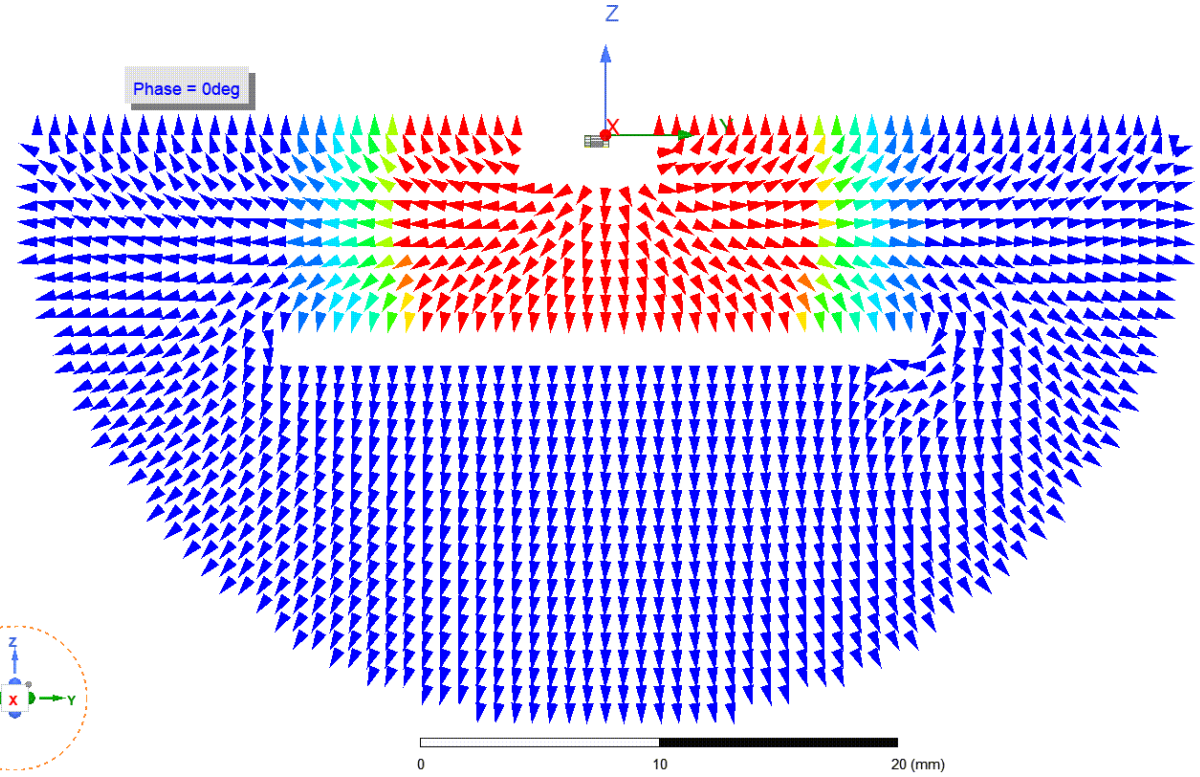




Standing Wave in inside cavity in cross direction

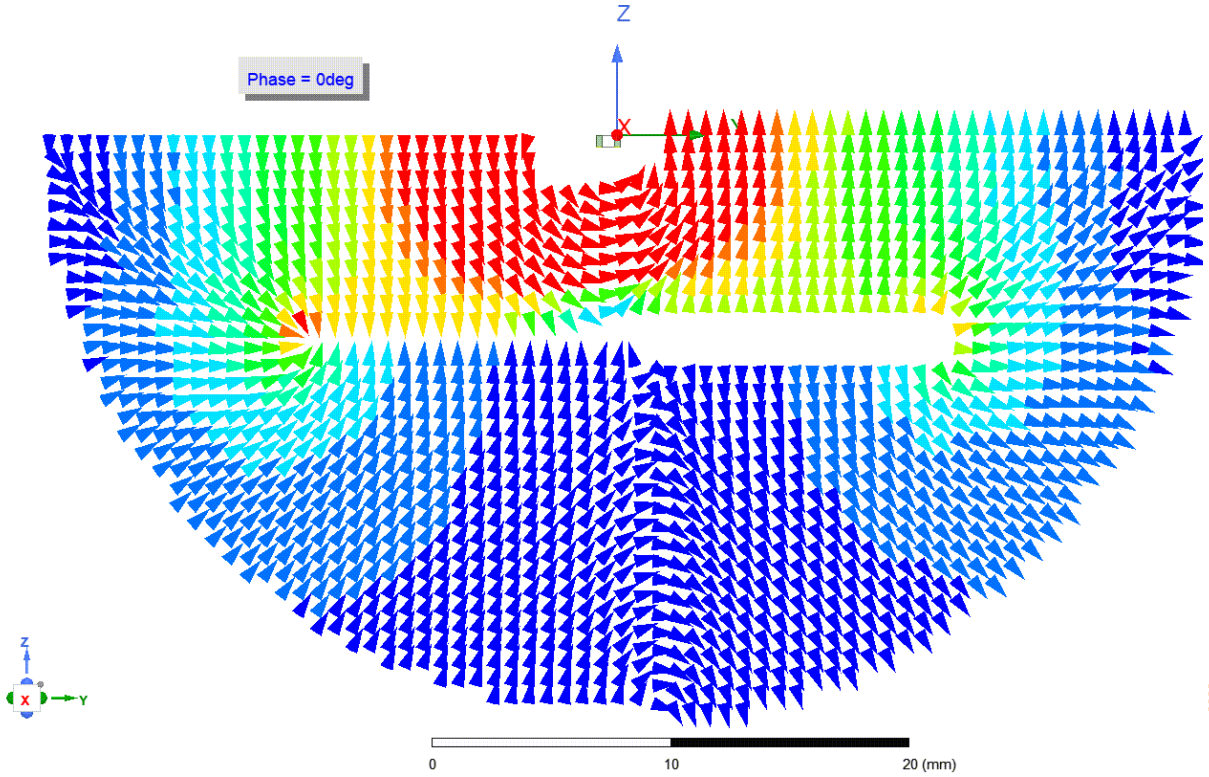


Electric Field at 2.3 GHz

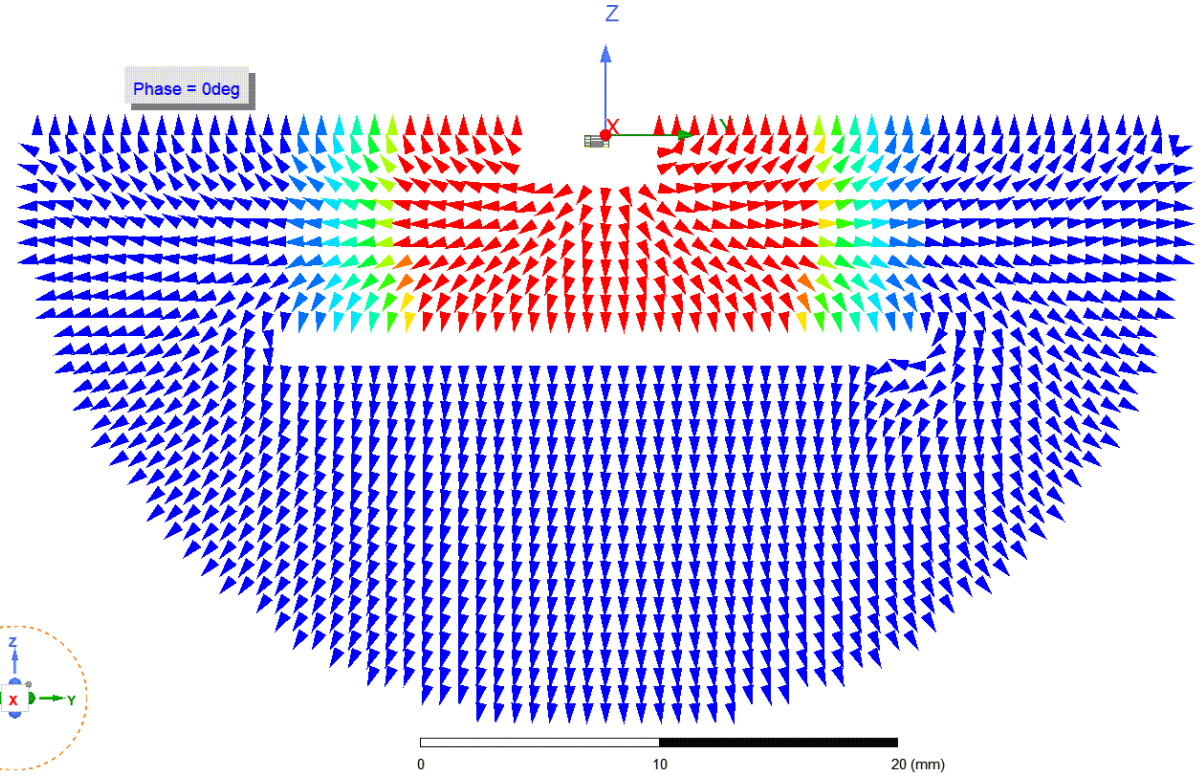


Electric Field at 100 MHz

Standing Wave in inside cavity in cross direction

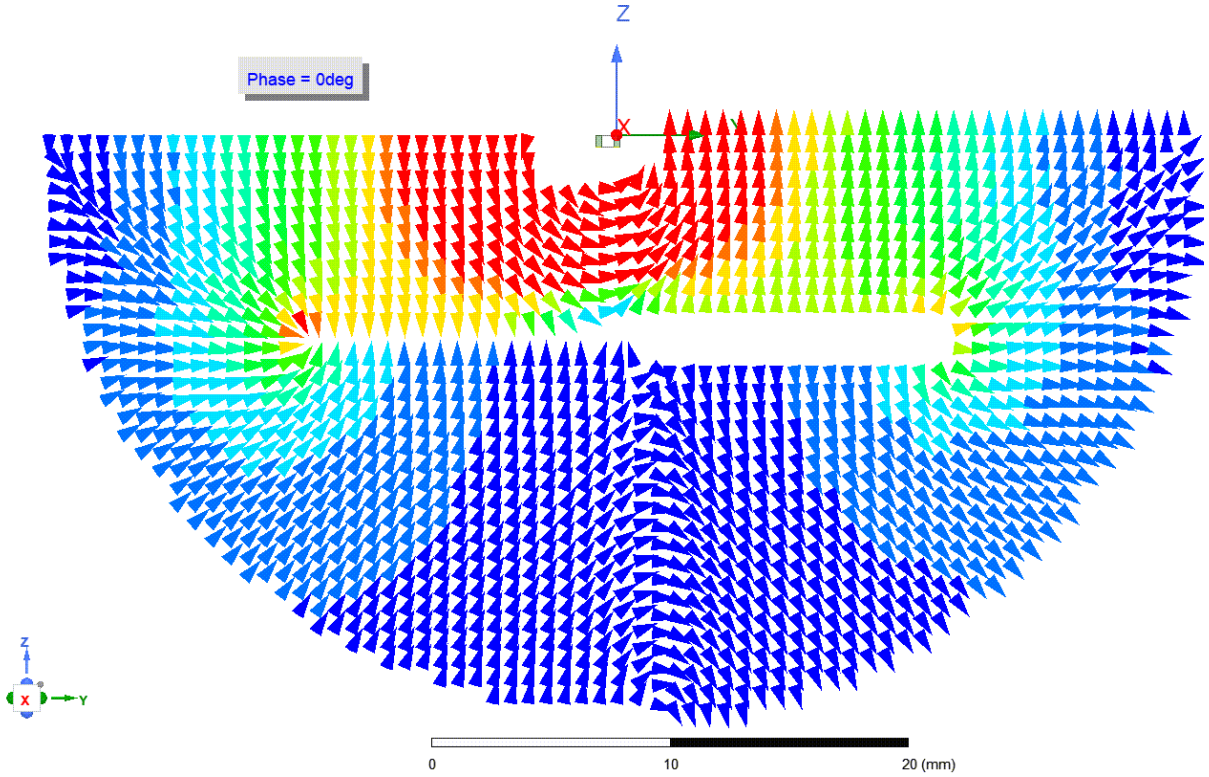


Electric Field at 2.3 GHz

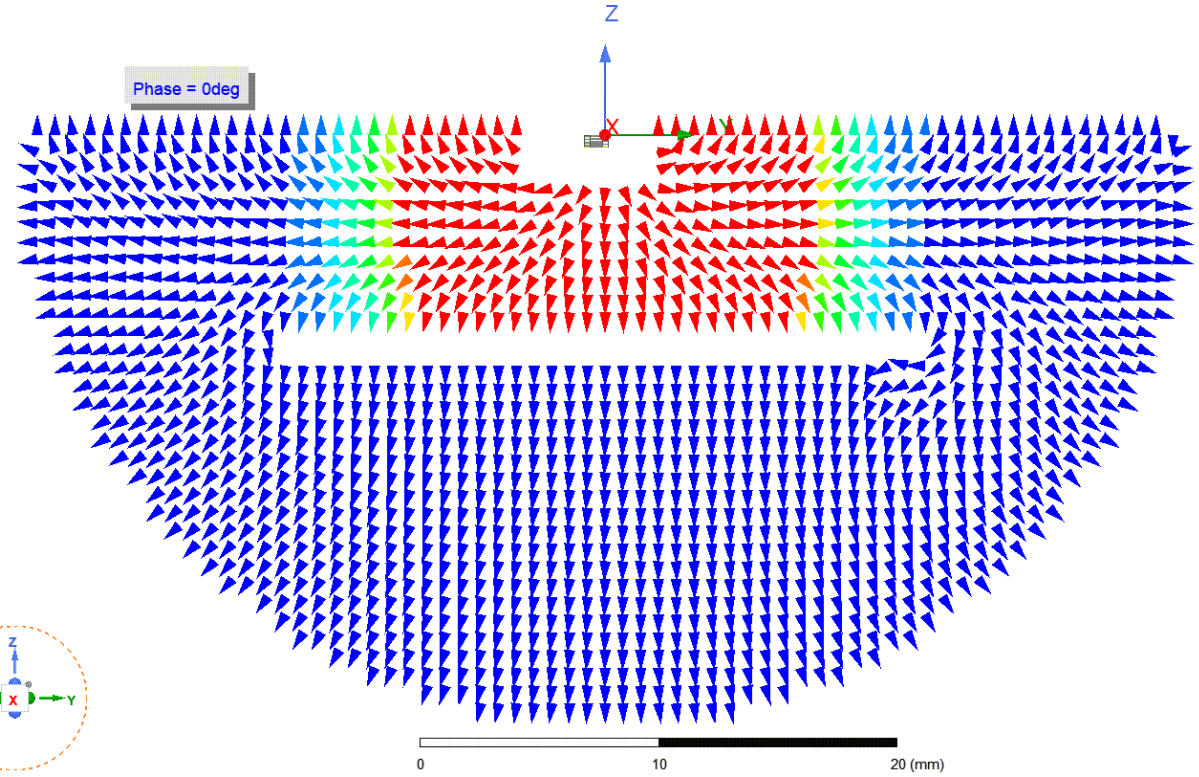


Electric Field at 100 MHz

Standing Wave in inside cavity in cross direction



Electric Field at 2.3 GHz



Electric Field at 100 MHz

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Conclusion and Outlook

- Near field scan experiments showed high potential emissions of buffer capacitors of optical laser emitters. The coupling behavior of MLCCs to a standardized radiated emission measurement system was investigated.

- It was shown that the orientation of the MLCC onside a PCB can have major influence on the coupling.

- By investigating the equivalent Hertzian dipole moments a possibility was found to distinguish between the two electrodes of a MLCC in the case of nonsymmetrical electrodes.

- Future studies will investigate simplified and numerically less expensive models for MLCCs and evaluate their benefits and limits with respect to radiated emissions.

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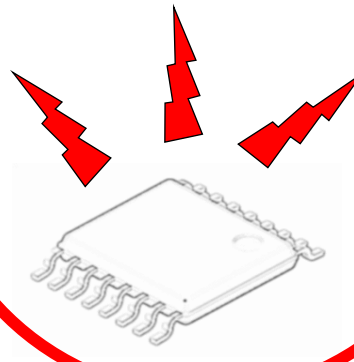
Thank you for your attention.

EMC (Electromagnetic Compatibility) on IC level

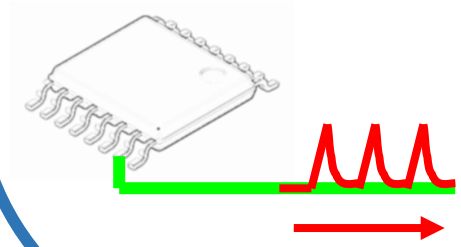
Electromagnetic Emission
IEC 61967

Electromagnetic Immunity (Susceptibility)
IEC 62132

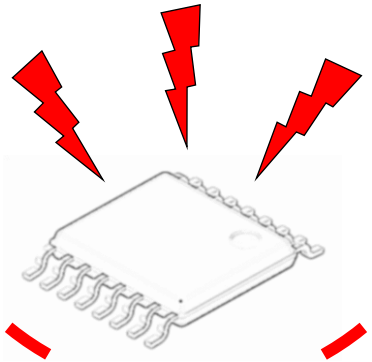
Radiated



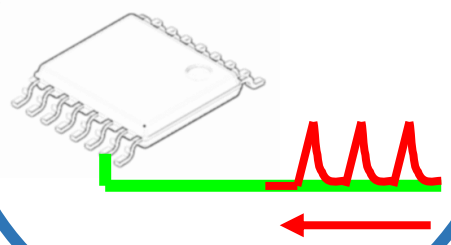
Conducted



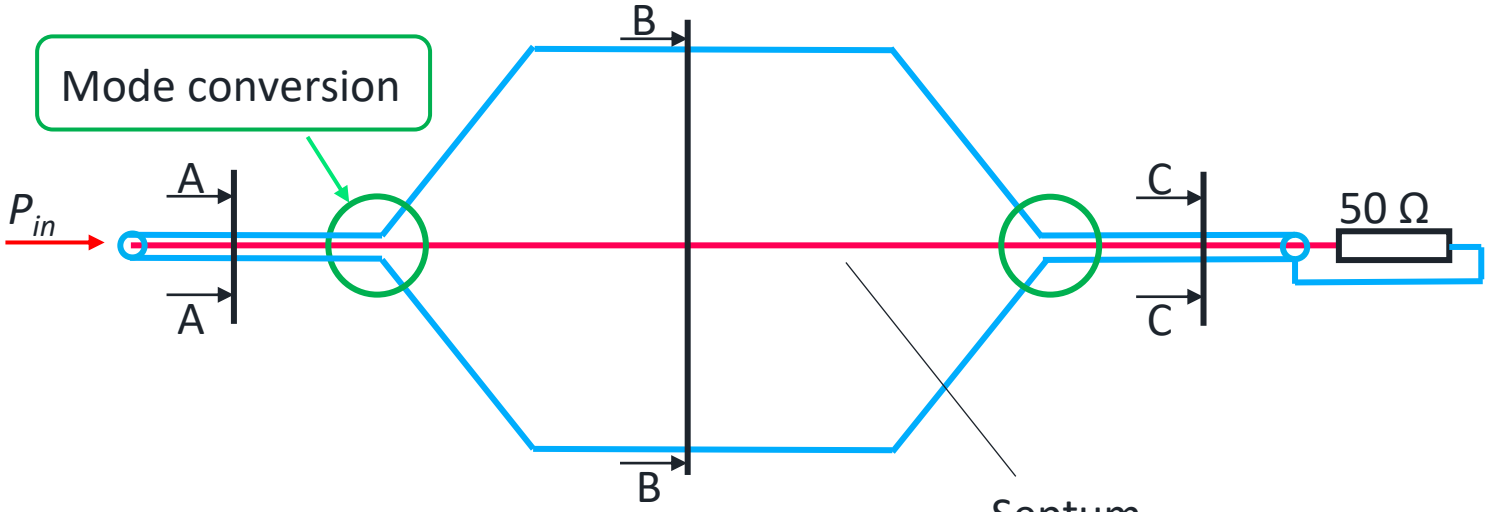
Radiated



Conducted

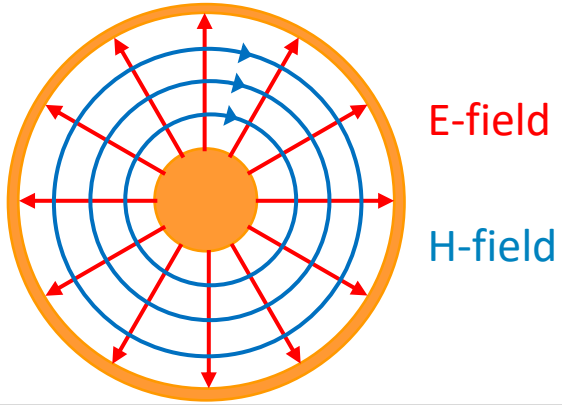


TEM cell method



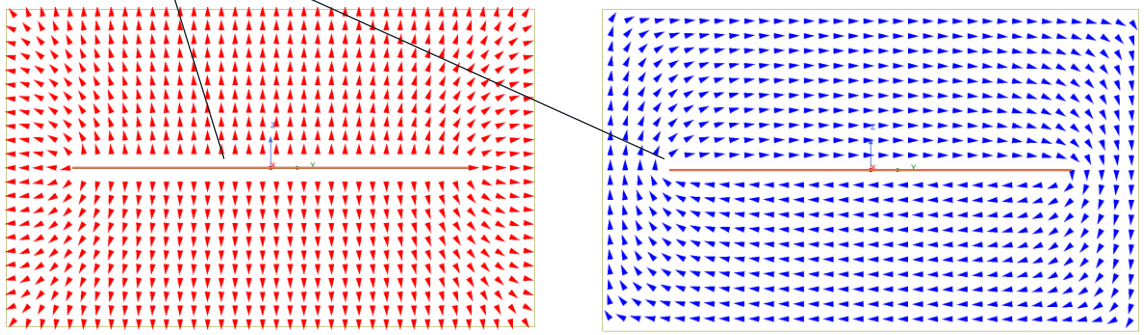
Mode conversion from cable to TEM cell: **coaxial** mode to **TEM** mode

Sectional view **A-A** and **C-C**:



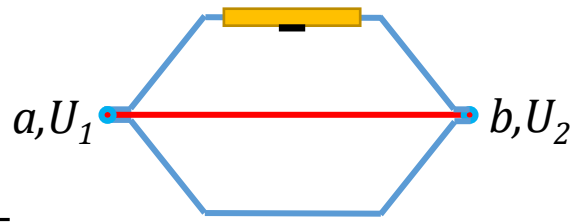
Septum

Sectional view **B-B**:



E-field H-field

Wave propagation inside TEM cell



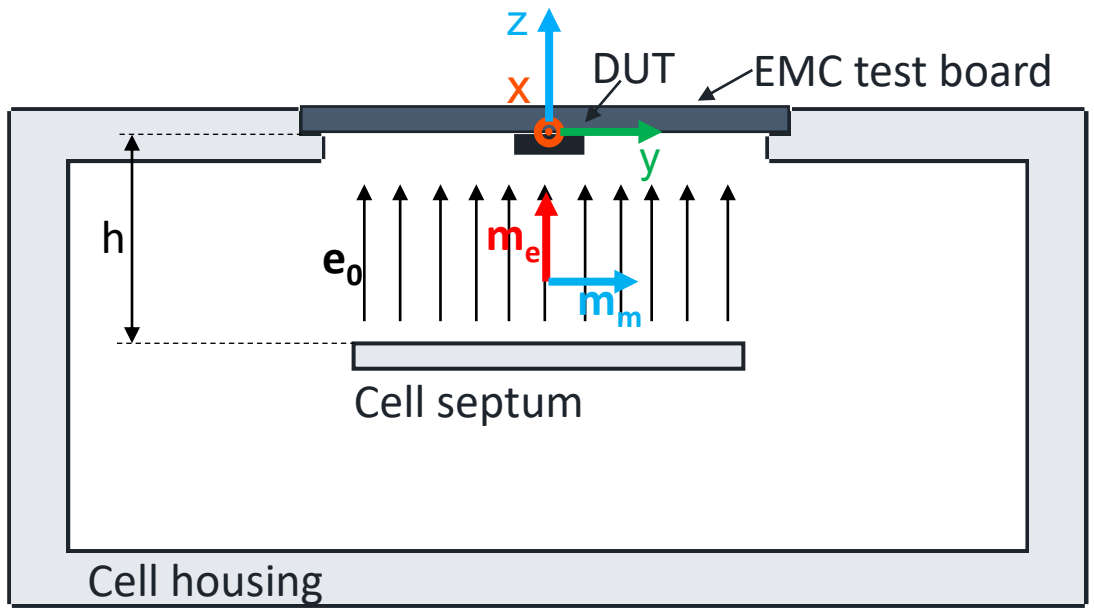
$$\binom{a}{b} = -\frac{1}{2}\left(\mathbf{m_e}\cdot\mathbf{e}_0 \pm jk\mathbf{M_m}\cdot(\mathbf{x}\times\mathbf{e}_0)\right) \text{ with } \mathbf{e}_0 = \left(\mathbf{z}e_{0z} + \mathbf{y}e_{0y}\right)[3], \quad e_{0z} \approx \frac{\sqrt{Z_W}}{h}, k = \frac{\mu_0\omega}{Z_0}, \mathbf{m_m} = \mathbf{M_m}\mu_0\omega$$

$$\binom{a}{b} = -\frac{1}{2}e_{0z}\left(m_{ez} \mp j\frac{m_{my}}{Z_0}\right), \qquad \binom{U_1}{U_2} = \binom{a}{b}\sqrt{Z_W}$$

$$\binom{U_1}{U_2} = -\frac{Z_W}{2h}\left(m_{ez} \mp j\frac{m_{my}}{Z_0}\right)$$

$$|m_{ez}| = \frac{|U_1 + U_2|h}{Z_W}, \qquad |m_{my}| = \frac{|U_1 - U_2|hZ_0}{Z_W}$$

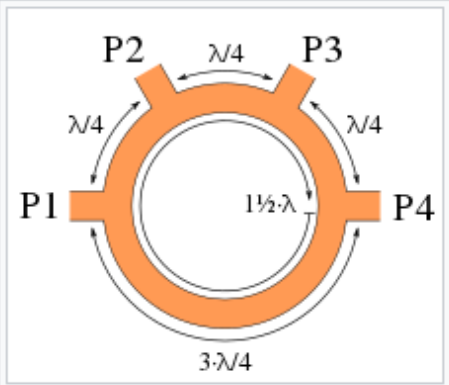
Z_W ... *Characteristic impedance*
Z_0 ... *Intrinisic wave impedance of free space*



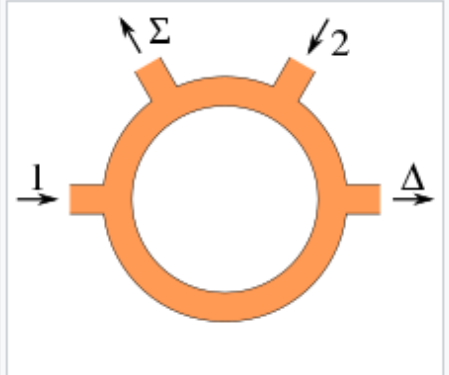
[3] I. Sreenivasiah, D. C. Chang and M. T. Ma, "Emission Characteristics of Electrically Small Radiating Sources from Tests Inside a TEM Cell," in IEEE Transactions on Electromagnetic Compatibility, vol. EMC-23, no. 3, pp. 113-121, Aug. 1981, doi: 10.1109/TEMC.1981.303930.

Rat race hybrid coupler

- Phase-correct addition, respectively subtraction of two input voltages
- Frequency range of this device: 1 GHz to 4 GHz, ultra wideband models not possible due to operating principle



Rat-race coupler



Arithmetics with rat-race coupler

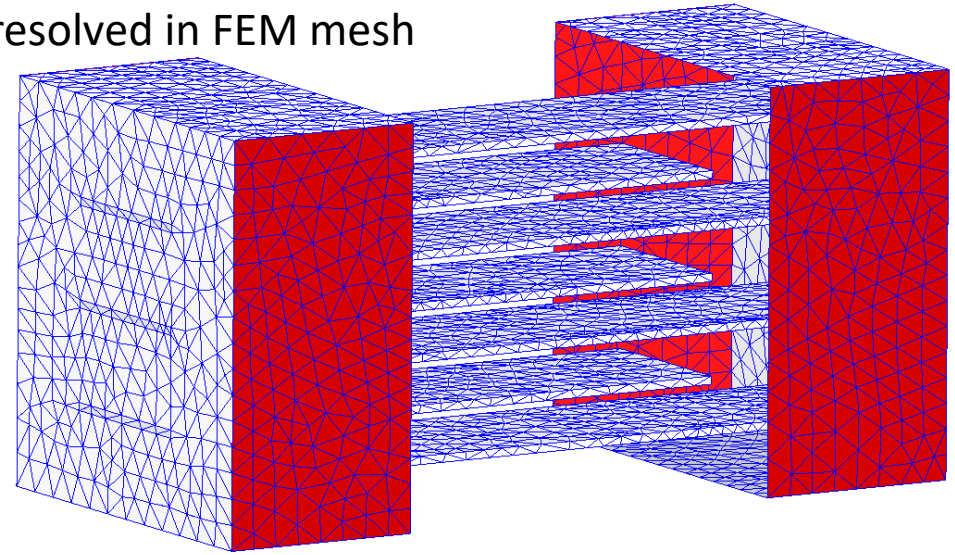
[5]

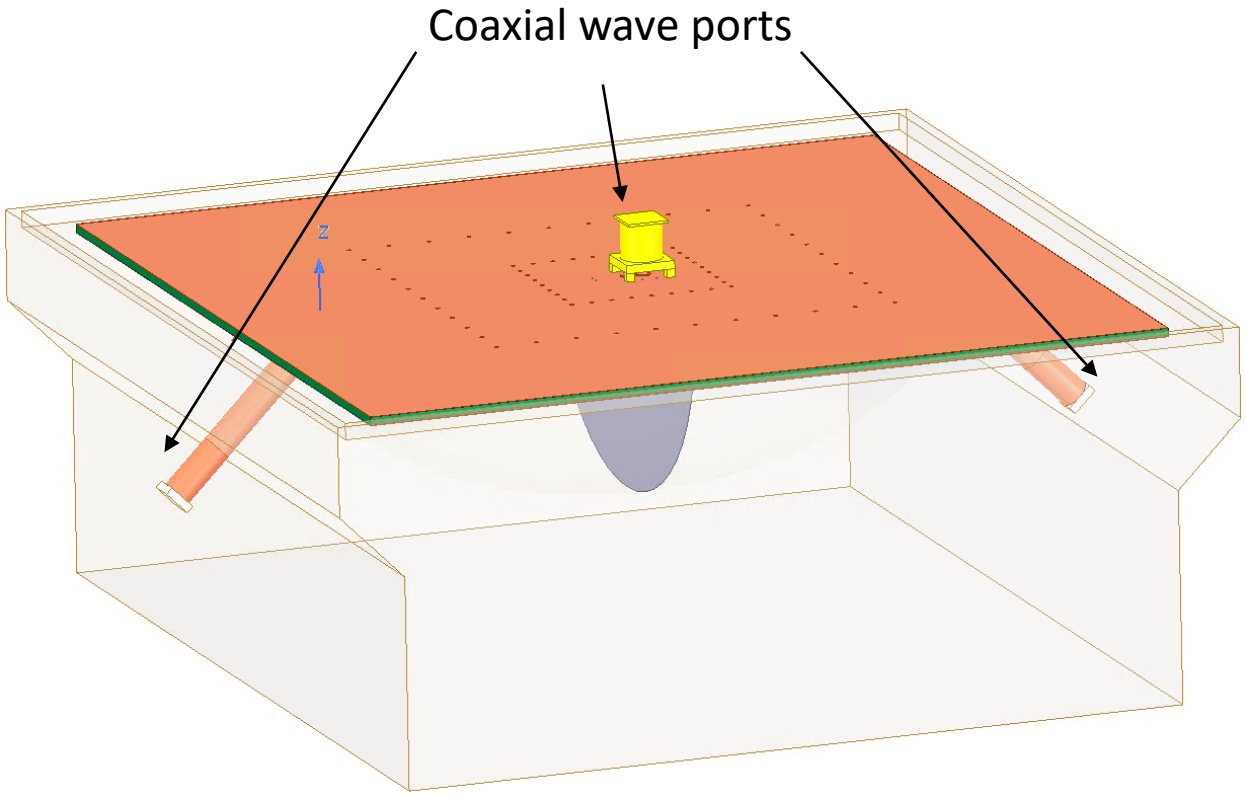
$$[\underline{S}] = \frac{-j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{bmatrix}$$

[5] https://en.wikipedia.org/wiki/Rat-race_coupler

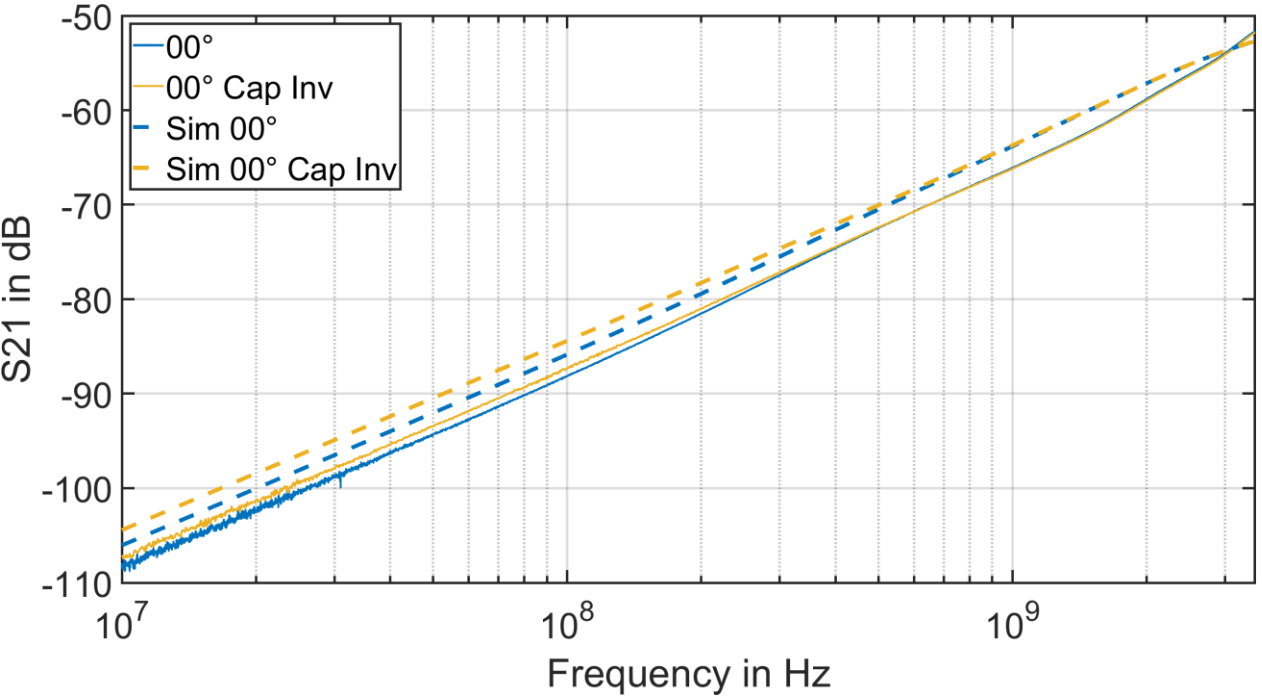
Simulation Setup

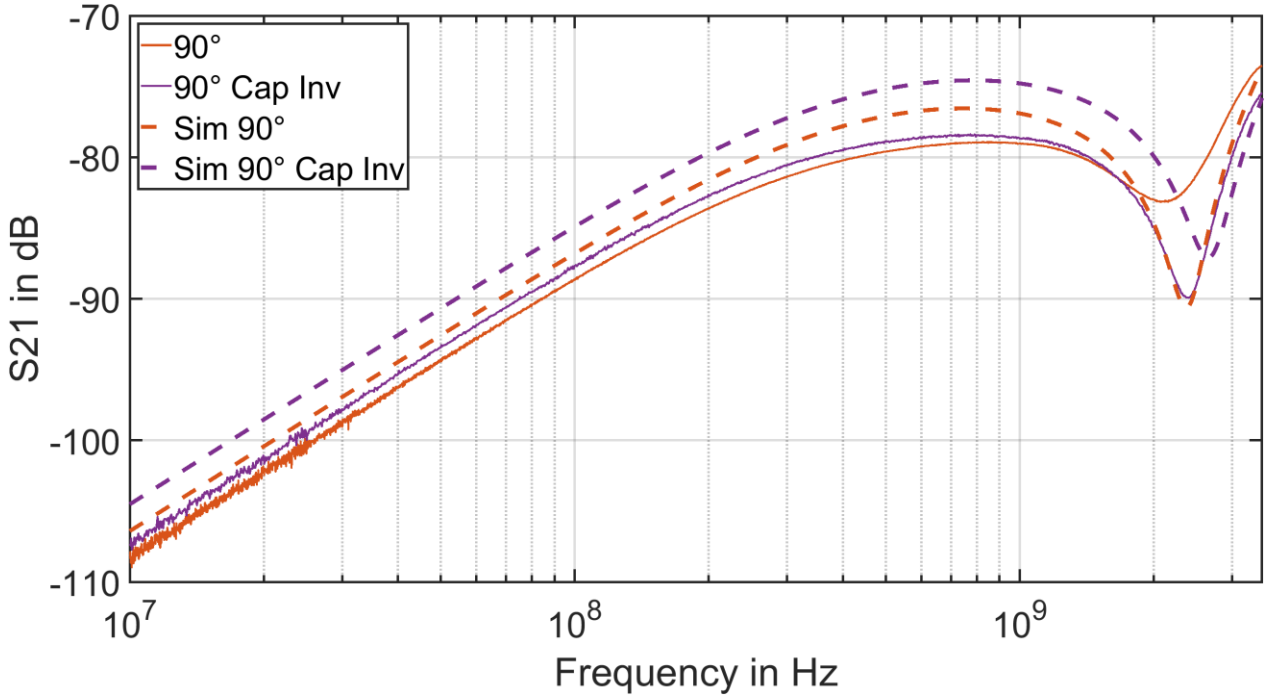
- A model of the exact measurement setup was created including IC stripline, PCB, connectors and DUT
- Wave ports used for S-parameter extraction
- Model of the internal finger structure of the PCB resolved in FEM mesh





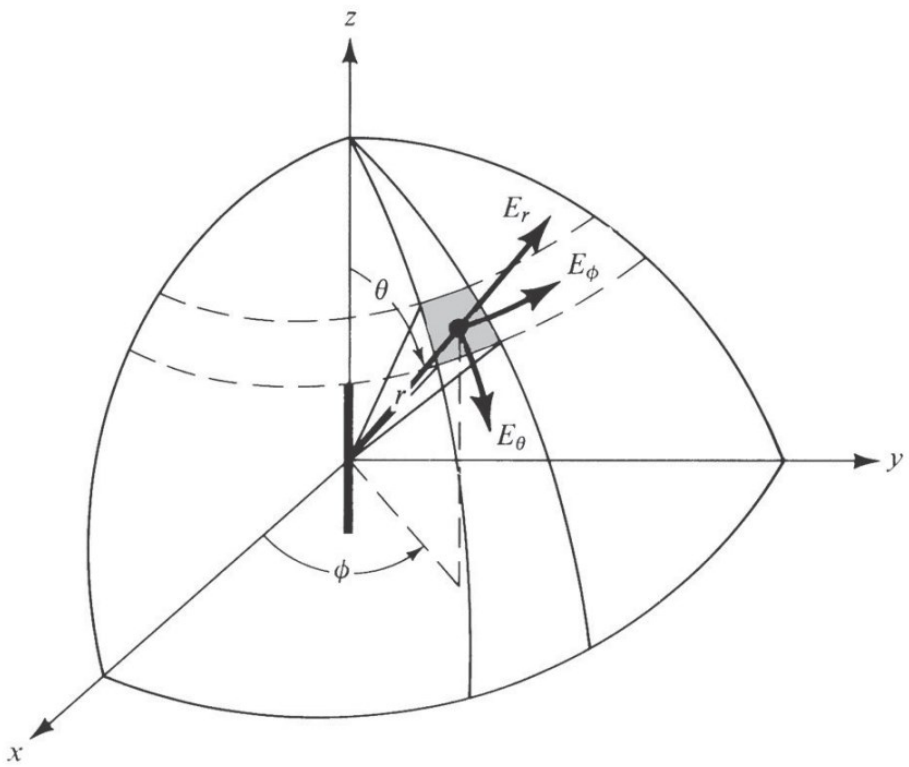
Results: Sim vs Meas





Incident Hertzian-Dipole Waves

Electric dipole



$$E_r = \eta \frac{I_0 l \cos\theta}{2\pi r^2}\left[1 + \frac{1}{jkr}\right] e^{-jkr}$$

$$E_\theta = j\eta \frac{k I_0 l \sin\theta}{4\pi r}\left[1 + \frac{1}{jkr} - \frac{1}{(kr)^2}\right] e^{-jkr}$$

$$E_\phi = 0$$

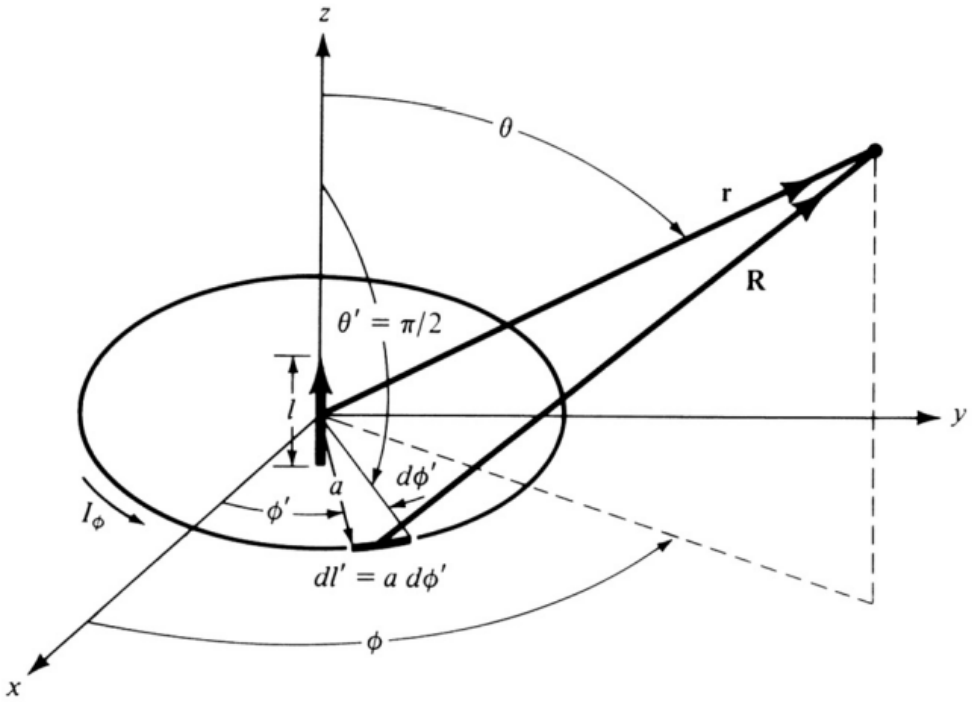
$$H_r = H_\theta = 0$$

$$H_\phi = j\frac{k I_0 l \sin\theta}{4\pi r}\left[1 + \frac{1}{jkr}\right] e^{-jkr}$$

[1] C. A. Balanis, Antenna Theory, Analysis and Design. WILEY Intersience, 2003.

Incident Hertzian-Dipole Waves

Magnetic dipole



$$E_r = E_\theta = H_\phi = 0$$

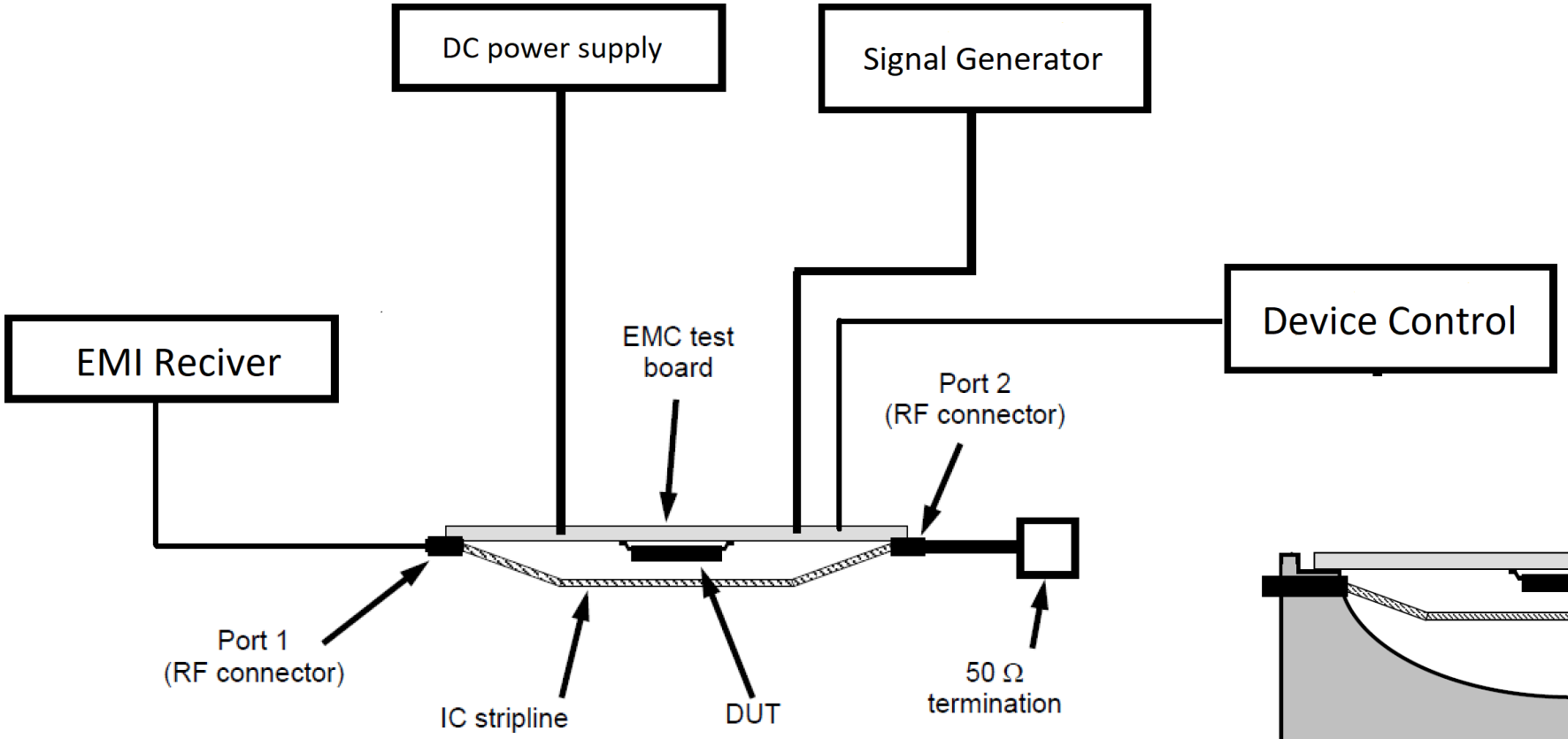
$$E_\phi = -j\frac{kI_m l \sin\theta}{4\pi r}\left[1 + \frac{1}{jkr}\right]e^{-jkr}$$

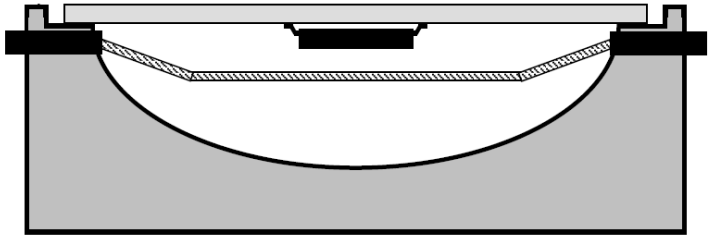
$$H_r = \frac{I_m l \cos\theta}{2\pi\eta r^2}\left[1 + \frac{1}{jkr}\right]e^{-jkr}$$

$$H_\theta = j\frac{kI_m l \sin\theta}{4\pi\eta r}\left[1 + \frac{1}{jkr} - \frac{1}{(kr)^2}\right]e^{-jkr}$$

[1] C. A. Balanis, Antenna Theory, Analysis and Design. WILEY Intersience, 2003.

Measurement Setup



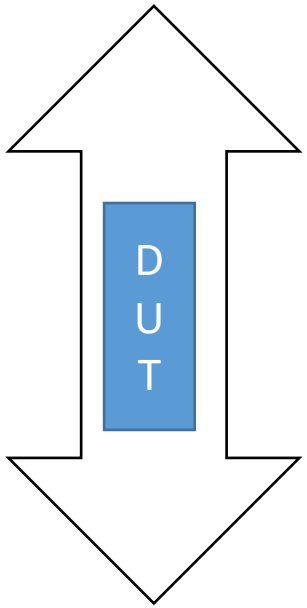


[6]

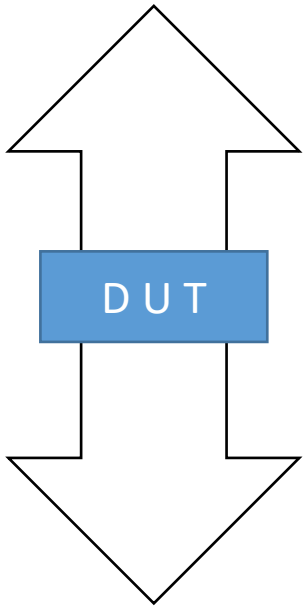
[6] IEC 61967; Integrated Circuits Measurement of Electromagnetic Emissions.

DUT orientations

0°

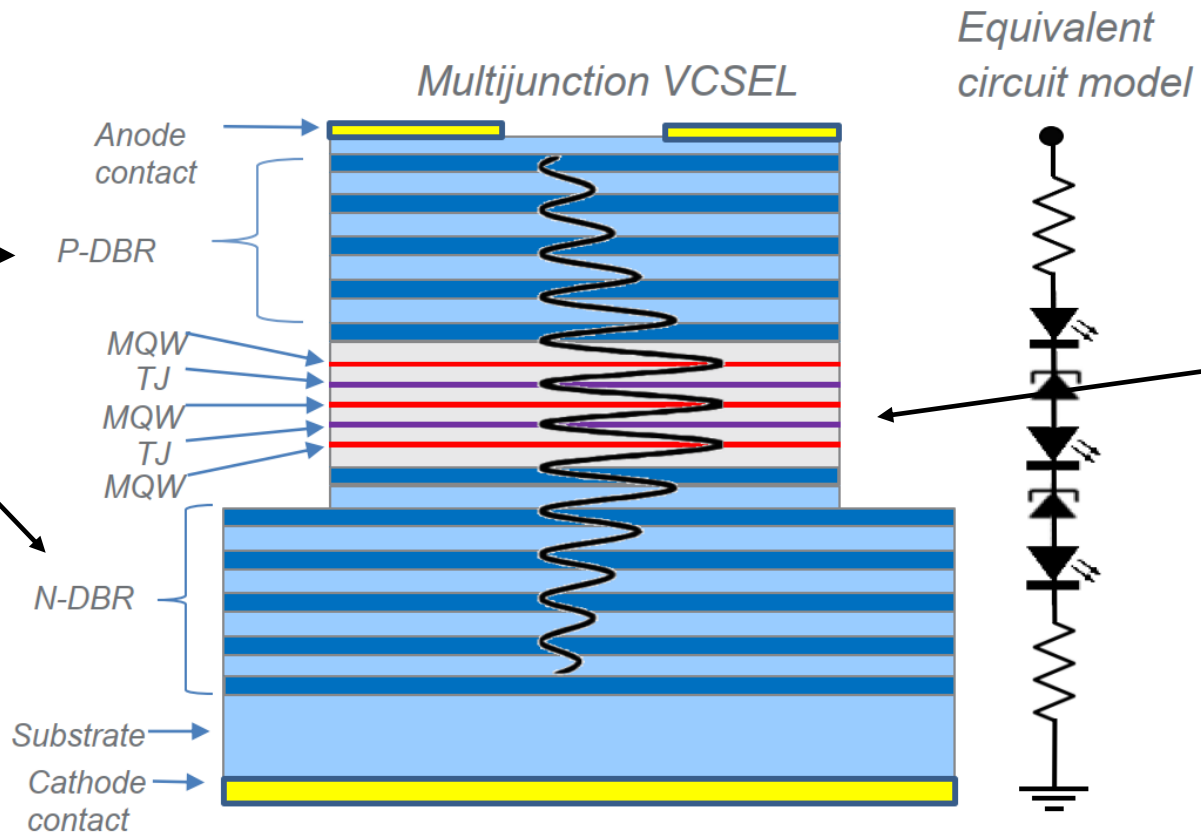


90°



Principle Sketch

Distributed Bragg Reflectors



Active area

[1] Vixar, ams OSRAM